

Selective Channel Invariance in Activity Recognition: Separating Static Multipath from Motion-Induced CSI Variations

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Abstract

WiFi signals can be used to recognize human activities like walking, sitting, or falling, without the cameras or wearable devices and are influenced by the environment through which they propagate. In an indoor setting, reflections from the furniture, walls, the static environment create multipath propagation paths between the transmitter and receiver. When a person moves through this static environment, these paths change producing variations in the measured CSI (channel state information) which are used for human activity recognition (HAR). However CSI also contains relatively stable patterns of the environment and when a training and testing data are collected in the same room, the models which are built may use these environment-specific channel patterns together with the changes caused by human activity movement. We call this as static environment learning bias where a model may just be memorizing the room, not the activity.

In this work, we study this problem in the UT HAR dataset and refer to the environment specific channel pattern as the room fingerprint. We examine whether the existing used WiFi HAR models rely on this fingerprint and whether this static environment learning can be reduced using data collected from a single room. For this, we use the physical characteristics of the CSI signal to suppress static environment channel components and retain discriminative variation associated with human motion. By introducing Selective Invariance Learning (SIL) which reduces the sensitivity of the learned representation to changes in the channel while preserving discriminative human activity recognition.

Our experiment provides consistent evidence that room specific information contributes substantially to the high accuracy obtained under the conventional single room evaluation. To prove the problem is real, we retrained a standard model BiLSTM using perturbed CSI on our processed data. The accuracy dropped from 96.2 % to 43.0%, which shows, it was heavily based on the room rather than the activity. Our SIL model, which was trained in the same way, achieved 77.8 % accuracy. Under a synthetic cross-room simulation, the baseline models fall to 16.0–22.7 % accuracy, whereas SIL retains 59.0%.

1. Introduction

1.1 Background

Wireless networks are traditionally built for communication- to send data from one device to another. They also interact with the physical environment while they travel. so they can be reused to sense that environment. This is known as Integrated Sensing and Communication (ISAC), where a single wireless system serves both purposes instead of depending on separate sensors (?).

WiFi-based Human Activity Recognition (HAR) is an application of this idea : It reuses the existing WiFi signals to detect activities like walking, sitting, or falling. This makes it useful in privacy-sensitive areas such as homes, hospitals, and elderly care.

1.2 Channel State Information (CSI)

In a WiFi system, the data is transmitted using OFDM, where the signal is split into many smaller frequency bands called subcarriers. When the signal travels from the transmitter to receiver, it reflects off the walls, furniture, and other objects. so the receiver receives multiple reflected copies combined together. This process is called multipath propagation.

Channel State Information (CSI) tells how this channel affects the signal. It records the change in amplitude and phase for each subcarrier. (?). CSI is sensitive - when the environment is still it stays roughly constant, but when the person moves the paths are disturbed so it changes. These changes are fed into a machine learning model which makes it possible to recognize the activities without a camera or wearable device.

1.3 The Room Fingerprint Problem

WiFi-based HAR models that are trained and evaluated on the standard benchmarks, report very high accuracy like 96.2%, 97.2%, and 87.0% for BiLSTM, CNN-GRU, and ViT respectively on UT-HAR (?). But, their data is collected from the *same room*. A room has a fixed layout so it creates constant reflections which produces a stable, room-specific pattern in the CSI signal - the **room fingerprint**. In both training and test data the fingerprint appears. so, the model recognizes the room not the activity which gives high accuracy. We call this the **fingerprint illusion**.



Figure 1: Two environments with identical transceiver placement and human walking path, but different room layouts which produces different multipath reflections. It creates geometry-dependent CSI pattern - the room fingerprint.

This matters because the benchmark accuracy may not guarantee real-world performance in a new room and calling it a "domain shift" doesn't explain *why* a model fails. Several Physical factors causes this: Constant reflections from walls, furniture, hardware effects like Automatic Gain Control (AGC), and slow environmental changes over time. All these are mixed into the same input, models can learn them along with genuine motion patterns. This similar to shortcut learning in deep learning (?). But here, the shortcut has a clear physical cause, which lets us design a solution based on the signal model itself.

1.4 Our Approach and Contributions

To address this problem , We propose **Selective Invariance Learning (SIL)**, which combines physics-informed signal preprocessing to reduce relatively static and unwanted channel components while retaining temporal variations associated with human motion with a contrastive training objective - VICReg (?) and Supervised Contrastive Learning (?). We evalaute SIL on UT-HAR using controlled gain perturbations and a synthetic cross room simulation.

The main contributions of this work:

- **The fingerprint problem:** With a Fingerprint-attenuated input, we retrain a standard BiLSTM model(?).The accuracy drops from 96.2% to 43.0%, which is almost as random guessing.
- **Selective Invariance Learning (SIL):** Our method removes the static components and trains the model to be selectively invariant to environmental change. On the same attenuated output,it achieves 77.8% accuracy.
- **Six diagnostic tests:** We use six complementary tests .These include embedding drift, cluster stability, label efficiency, per-class accuracy breakdown, simulated cross-room test, and collapse test. In the simulated cross-room test, SIL obtains 59.0% accuracy, comapred

with the standard baseline models achieving only 16.0-22.7%.

- **Controlled ablation study:** The two components of SIL are signal preprocessing and a training method. We test all four combinations with and without each to understand their contribution.
- **Reusable protocol:** We combine the six diagnostics into an environmental dependence evaluation protocol that can be applied directly to single-room WiFi HAR datasets, without requiring real multi-room data.

2. Related Work

2.1 WiFi CSI Activity Recognition

WiFi signals at first used with handcrafted features (??), and later with deep learning models like CNNs, RNNs, and transformers, which increased accuracy above 90% (??). But most models are trained and tested in the same room, so the accuracy may fall in a new room.

2.2 Cross-Environment Generalization

Some methods try to fix cross-room performance directly. Widar3.0 (?) uses velocity features which doesn't depend on the room, but needs special hardware. CrossSense (?) and EI (?) train on data from multiple rooms. Similar ideas exist in RF sensing more broadly (?). Our method, SIL, is different: it removes room-specific information at the source. As we are using only single-room data. Whether it works on real unseen rooms is still open.

2.3 Spurious Feature Learning

Geirhos et al. (?) shows that models often learn shortcut features instead of real patterns, which may fail later during deployment. Our fingerprint problem is similar to this. The model uses the room CSI patterns as shortcuts instead of learning the activity. But in our case, the spurious information has physical basis (Eq. 1), so we can fix it using the actual signal model.

2.4 Invariant and Contrastive Representation Learning

Invariant Risk Minimization reduces dependence on environment-specific features, but it needs data from multiple environments (?). With VICReg the representations are kept stable through variance and covariance regularization, without using negative pairs(?). Supervised Contrastive Learning (?) groups similar activities using class labels. SIL combines both VICReg and SupCon with a physics-based gain augmentation step. To the best of our knowledge this hasn't been done before for single-room CSI data.

3. Methodology: Selective Invariance Learning

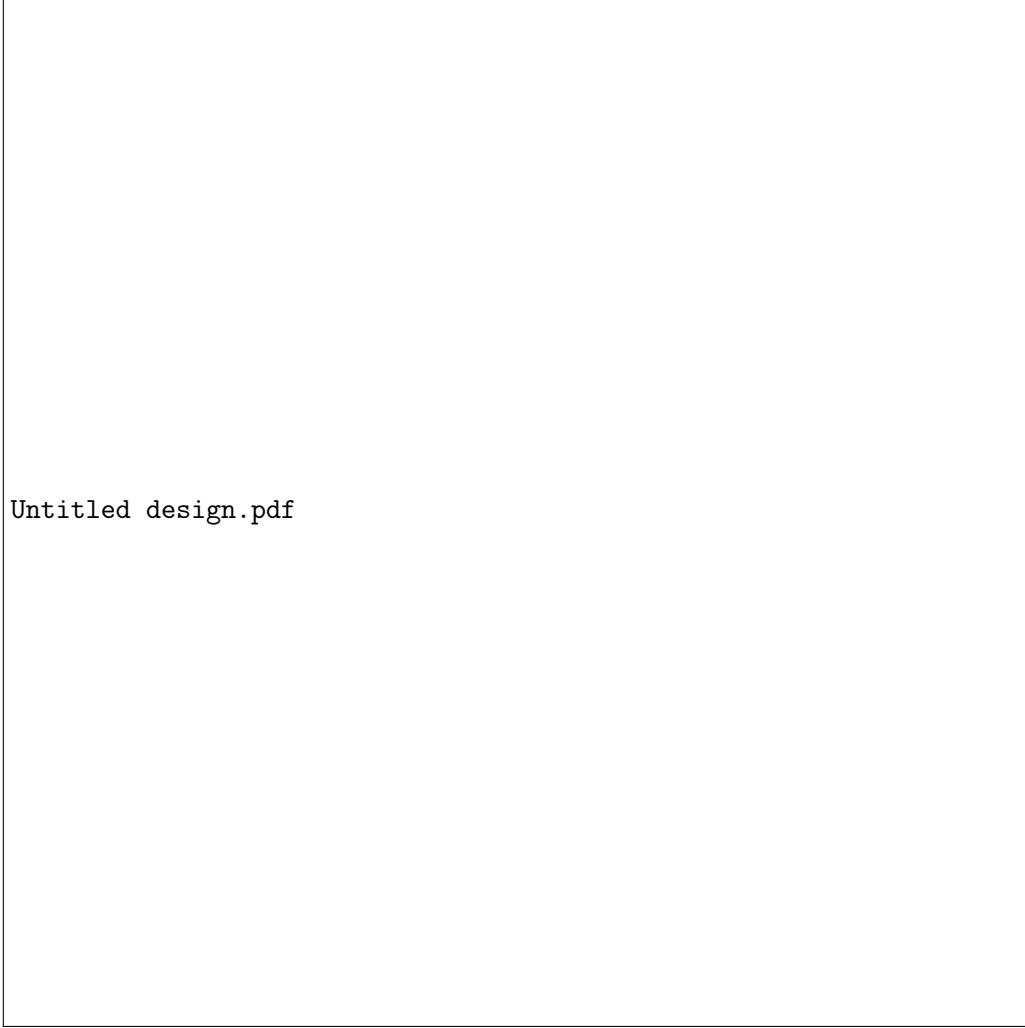


Figure 2: Overview of the Selective Invariance Learning (SIL) pipeline, from raw CSI input through preprocessing, encoding, and training.

3.1 The Fingerprint Problem, Mathematically

The CSI signal can be split into two parts: a fixed, room-specific component, and a smaller component caused by motion:

$$H(t, k) = \underbrace{H_{\text{static}}(k)}_{\text{room fingerprint}} + \underbrace{\sum_{m \in \mathcal{M}(t)} a_m(t) e^{-j2\pi f_c \tau_m(t)}}_{\text{motion-induced component}} \quad (1)$$

where $\mathcal{M}(t)$ is the set of moving-body propagation paths, a_m and τ_m are complex attenuation and delay, and f_c is the carrier frequency. The static term is much larger than the motion term, $|H_{\text{static}}(k)| \gg |H_{\text{motion}}(t, k)|$ in typical indoor settings. So, the magnitude of actual motion signal $|H(t, k)|$ is dominated by the static term.

For cross-room deployment, the static response shifts to $H'_{\text{static}}(k) = (1 + g_k)H_{\text{static}}(k)$ where $g_k \sim \mathcal{N}(0, \sigma_{\text{gain}}^2)$ encodes per-subcarrier frequency-selective fading from new room geometry. The model encodes $H_{\text{static}}(k)$ as a discriminative feature and when placed in a new room it encounters a different input. This is called distribution shift. We operationalize the gain-perturbation component of room change as independent per-subcarrier multiplicative noise:

$\tilde{X}(t, k) = (1 + g_k)X(t, k)$, $g_k \sim \mathcal{N}(0, 0.4^2)$. Important distinction: in evaluations of baseline models (BiLSTM, CNN-GRU, ViT), this perturbation is applied to raw CSI amplitude and primarily disrupts the static fingerprint component $H_{\text{static}}(k)$ that dominates the amplitude. In evaluations of SIL, gain perturbation is applied to SIL-preprocessed input, from which the temporal mean has already been subtracted (Stage 2); it therefore tests residual gain sensitivity in the motion-band signal, not fingerprint disruption. These two settings measure different things and are not directly comparable. The synthetic model is calibrated to inter-session amplitude variability in UT-HAR and is a deliberate simplification; real cross-room channels additionally exhibit correlated fading and delay spread changes (Section 4.10).

3.2 Preprocessing: Removing the Room Fingerprint

SIL applies a five-stage preprocessing pipeline before the signal reaches the model:

1. **Outlier removal** - a Hampel filter suppresses hardware glitches from automatic gain control.
2. **Static detrending** - the per-subcarrier time-average is subtracted, removing the room fingerprint:

$$X_d(t, k) = X_h(t, k) - \frac{1}{T} \sum_{\tau=1}^T X_h(\tau, k) \quad (2)$$

3. **Motion-band filtering** - a bandpass filter (0.5–20 Hz) retains motion-relevant frequencies and discards slow drift and fast noise.
4. **Normalization** - per-subcarrier Z-score normalization using training statistics.
5. **Velocity/acceleration enrichment** - temporal derivatives are appended to capture how the signal changes over time.

The result is converted into a spectrogram, which is the actual input to our model.

3.3 Model Architecture

The preprocessed spectrogram passes through three convolutional blocks, followed by multi-head self-attention, producing a 256-dimensional embedding. A linear layer maps this embedding to one of seven activity classes.

3.4 Training Objective

SIL is trained using a combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda_{\text{inv}} \mathcal{L}_{\text{VICReg}}(z, \tilde{z}) + \lambda_{\text{sc}} \mathcal{L}_{\text{SupCon}}(z, y) \quad (3)$$

$\mathcal{L}_{\text{CE}}$ trains the model to classify activities correctly. $\mathcal{L}_{\text{VICReg}}$ (?) encourages the model to produce similar embeddings before and after a gain perturbation, teaching it to ignore environment-driven changes. $\mathcal{L}_{\text{SupCon}}$ (?) pulls embeddings of the same activity class together, so the invariance objective does not come at the cost of losing class separability. The model is trained using AdamW with cosine annealing, for up to 150 epochs with early stopping, and the VICReg term activates after a 5-epoch warmup.

4. Experimental Findings

Dataset and Protocol. All experiments are conducted on the UT-HAR dataset (??) which contains seven activities: Lying, Sitting, Standing, Walking, Running, Falling, and Walking Fast, with a total of 4,973 samples. A fixed test set of 500 samples is used for evaluation. Baselines (BiLSTM, CNN-GRU, ViT) are trained on raw amplitude following the SenseFi protocol (?). Unless otehrwise stated , experiment use the same random seed $s = 42$. The main effects

including the 53.2 pp collapse and $2.9\times$ drift ratio are consistent across different random seeds. Smaller differences of around 1–3 pp should be treated as indicative, as they may reflect variation across experimental runs. The exception is the 2×2 ablation study where (A1, B1) cells are evaluated over 5 seeds and are reported as mean $\pm$ std. *All perturbations including the synthetic cross-room simulation are applied in-dataset; no real multi-room CSI measurements are used.*

4.1 Diagnostic 1 - Representation Drift Under Gain Perturbation

We compare the encoder representations of clean and gain-perturbed CSI samples while keeping the encoder fixed, the embeddings representations are first L2 normalised and their average Euclidean distance is used to measure the representation drift. A lower value indicates that the drift is less affected by the channel perturbations:¹

$$\Delta L_2 = \frac{1}{N} \sum_{i=1}^N \|z_i^{\text{clean}} - z_i^{\text{perturbed}}\|_2 \quad (4)$$

Table 1: Embedding drift (ΔL_2 , lower = more invariant) under five perturbation types. All embeddings are L2-normalized to before computing drift to ensure comparability across architectures. Fix seed s=42 is used for all perturbation experiments

Perturbation	SIL (Proposed)	BiLSTM	CNN-GRU	ViT
Gain mismatch ($1\times$, $\sigma=0.4$)	0.275	0.312	0.807	0.601
Gain mismatch ($2\times$, $\sigma=0.8$)	0.425	0.659	1.094	0.970
Gain mismatch ($3\times$, $\sigma=1.2$)	0.463	0.871	1.223	1.128
SNR fading noise	0.357	0.272	0.215	0.250
Background drift	0.176	0.148	0.414	0.280

Table ?? and Figure ?? show that CNN-GRU’s gain drift (0.807 at $1\times$) is $2.9\times$ higher than SIL’s (0.275), growing to 1.223 at $3\times$ while SIL reaches only 0.463. This indicates that SIL is more stable under the gain variations used during training. CNN-GRU is more stable under SNR fading. SIL is not simply less sensitive to all perturbations, it is specifically insensitive to the perturbations it was trained to handle. SIL also shows lowest background-drift (0.176), consistent with DC-stripping preprocessing reducing sensitivity to slow environmental drift.

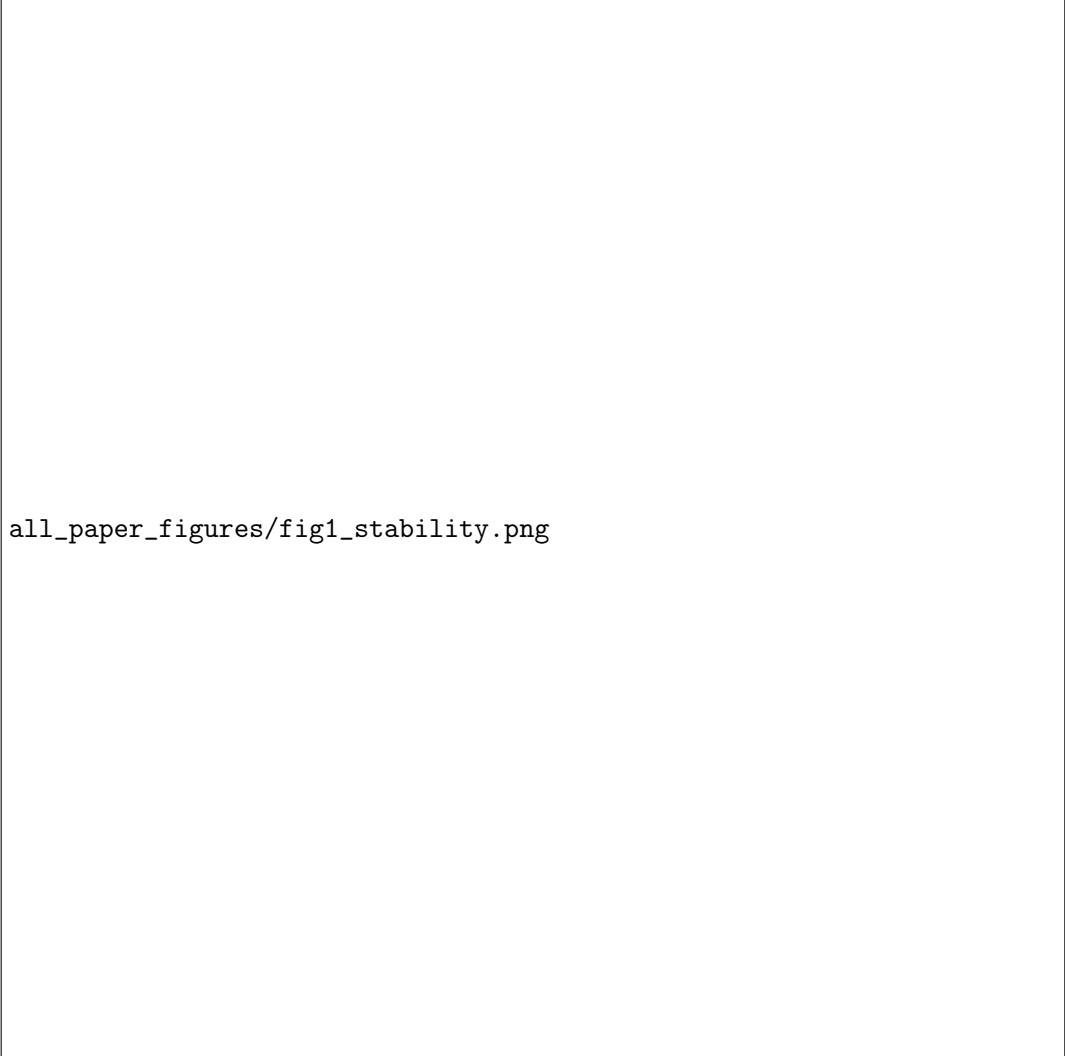
4.2 Diagnostic 2 - t-SNE Cluster Stability

We apply t-SNE (perplexity 30, PCA initialization, 1000 iterations, seed 42) both to clean and gain@ $2\times$ embeddings of SIL and CNN-GRU, to visualise how their embeddings change under channel perturbations. Using the same t-SNE model for both, CNN-GRU shows a centroid shift of 13.34 units, which is $2.66\times$ larger than SIL’s shift of 5.01 units, consistent with the $2.9\times$ L2 indicating greater representation change under the gain perturbation. This is consistent with the higher L2 drift observed for CNN-GRU. The clean CNN-GRU embeddings also form a compact tight cluster than SIL. While this compact representation gives high classification accuracy, it may indicate that the model is learning the static environment specific patterns present in the samples.

4.3 Diagnostic 3 - Label Efficiency Probing

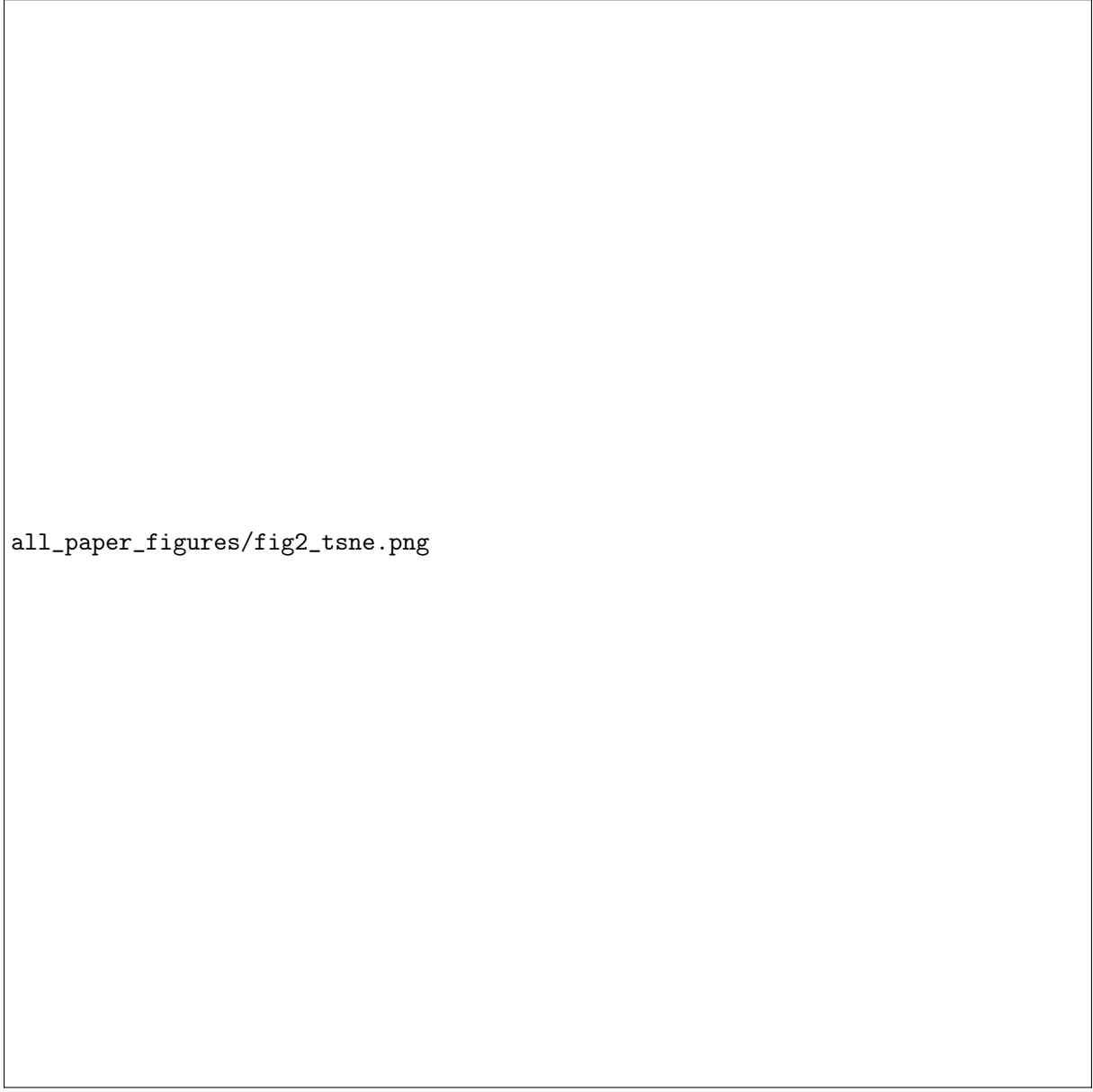
We freeze each encoder and train a logistic regression probe using different parts of training data ranging from 1% (≈ 40 samples) to 100%.

¹All embeddings are L2-normalized to the unit hypersphere, ensuring drift magnitudes are comparable across architectures of different capacity.



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Figure 3: Embedding drift per model per perturbation type. SIL achieves the lowest drift on all gain-related perturbations ($1\times-3\times$). Under SNR fading, CNN-GRU shows the lowest drift (0.215), while SIL is highest (0.357). This is because SIL's temporal-derivative features emphasize rapid CSI changes, making the representation more sensitive to high-frequency noise. Thus, SIL's invariance is targeted toward the channel variations considered during training rather than being invariant to all perturbations.



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Figure 4: t-SNE projections of test-set embeddings under clean (left) and gain@2× perturbed (right) conditions. Top: SIL. Bottom: CNN-GRU. CNN-GRU forms near-perfect clusters on clean data, then shows centroid displacement under gain perturbation (mean shift: 13.34 units). SIL clusters are less compact on clean data but remain stable under perturbation (shift: 5.01 units) - a 2.66× smaller displacement, consistent with Table ?? . Although, CNN-GRU has more compact clusters under clean data, it reflects its dependence on static environment invariant information; not activity-level variation.

Table 2: Frozen encoder linear probe accuracy at three labeled data fractions (%).Rapid saturation suggests easily separable representations at low label fractions , while one requiring discriminative activity information would improve more gradually.

Label fraction	SIL (Proposed)	BiLSTM	CNN-GRU	ViT
1% (≈ 40 samples)	76.1 ± 1.4	94.2 ± 1.5	97.1 ± 0.2	81.2 ± 3.5
10% (≈ 398 samples)	77.3 ± 0.8	96.2 ± 0.2	97.3 ± 0.1	88.9 ± 0.4
100% (full training set)	77.4	96.2	97.2	89.4

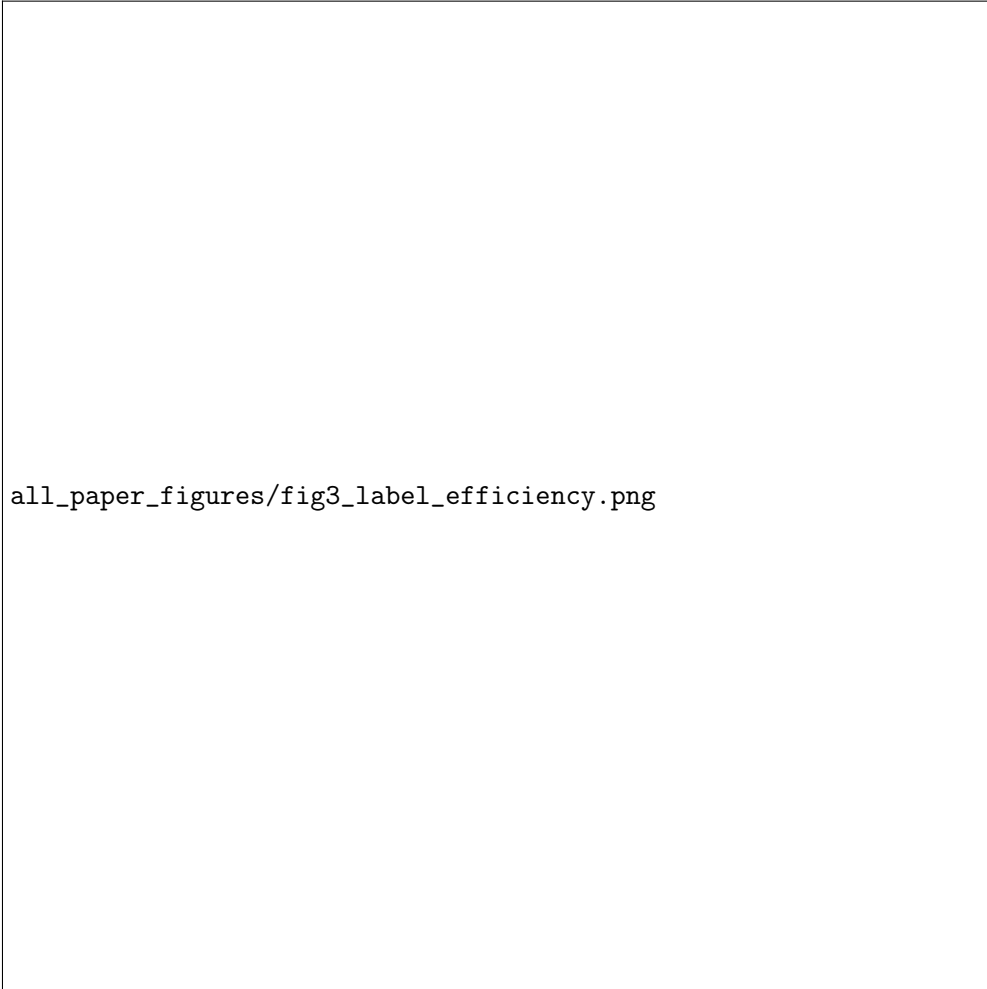


Figure 5: Linear probe accuracy vs. labeled training fraction. CNN-GRU reaches 97.1% with just 40 samples, and improves by just 0.1 pp from the remaining 3,937 data, indicating rapid saturation. Contrary, SIL's gradual improvement (76.1% $\rightarrow$ 77.4%) reflects genuine activity level learning that requires class supervision to resolve. Since the evaluation uses a single environment, this result is interpreted with the drift and collapse analyses together.

CNN-GRU achieves 97.1% probe accuracy with 40 labeled training samples (1% data) matching its fully supervised performance of 97.1%. BiLSTM similarly reaches about 94% at 1%. SIL improves gradually from 76.1% to 77.4%, a pattern consistent with representations that require genuine labeling to resolve class boundaries., this suggests that the SIL representation relies more on discriminative activity related information, whereas the rapid saturation of the baseline representations may indicate that they contain easily separable static information. Since the dataset uses a single environment, this result is interpreted alongside the drift and per-class analyses.

4.4 Diagnostic 4 - Per-Class Dynamic Activity Breakdown

We divide the seven activities into static (Lying, Sitting, Standing; 258 test samples) and dynamic activity classes (Walking, Running, Falling, Walking Fast; 242 test samples). All group accuracy values in Table ?? are *sample-weighted* (total correct within the group / total samples in the group), to maintain consistency with the overall accuracy . Table ?? and Figure ?? reveal a

Table 3: Sample-weighted group accuracy (%) at three gain perturbation levels. *Drop* column: clean $\rightarrow$ gain@2 $\times$ decline in percentage points. All values satisfy: $\min(\text{Static}, \text{Dynamic}) \leq \text{Overall} \leq \max(\text{Static}, \text{Dynamic})$. Perturbation draws use seed 42; values vary by ± 3 pp across independent seeds.

Model	Group	Accuracy (%)			Drop (pp)
		Clean	Gain@1 $\times$	Gain@2 $\times$	
SIL (Proposed)	Static	76.4	63.2	48.4	28.0
	Dynamic	79.3	68.6	53.3	26.0
	Overall	77.8	65.8	50.8	27.0
BiLSTM	Static	97.3	91.5	69.0	28.3
	Dynamic	95.0	86.8	66.9	28.1
	Overall	96.2	89.2	68.0	28.2
CNN-GRU	Static	98.1	76.4	47.3	50.8
	Dynamic	96.3	74.8	42.6	53.7
	Overall	97.2	75.6	45.0	52.2
ViT	Static	91.1	68.6	44.6	46.5
	Dynamic	82.6	55.0	32.2	50.4
	Overall	87.0	62.0	38.6	48.4

qualitatively important distinction. CNN-GRU and ViT show markedly larger drops on dynamic classes (Δ_{drop} : CNN-GRU +2.9 pp, ViT +3.9 pp), consistent with the fingerprint thesis: static posture activities retain some geometry-specific reflection patterns under perturbation, while dynamic motion activities which should be the *primary* target of a HAR system collapse most sharply. SIL’s nearly symmetric drops (−2.0 pp static vs. dynamic) indicate uniform residual gain sensitivity that is not class-specific and is not fingerprint reliance.

BiLSTM’s overall and dynamic drops (28.2 pp and 28.1 pp) are comparable to SIL’s (27.0 pp and 26.0 pp), suggesting some inherent temporal stability. However, BiLSTM’s embedding drift grows substantially with perturbation intensity (0.312 at 1 $\times$ to 0.871 at 3 $\times$) while SIL’s grows more slowly (0.275 to 0.463) and the collapse test (Section ??) reveals the ultimate nature of BiLSTM’s apparent robustness.

4.5 Clean Accuracy and 5-Fold Cross-Validation

SIL achieves 77.8% clean accuracy on the fixed test split, with a test F1-score of 74.0%. Under 5-fold cross-validation, it achieves a mean validation accuracy of $83.3\% \pm 0.92\%$ and a test accuracy of 79.6%. This shows that the performance is reasonably stable across different data partitions. The 19.4 pp gap to CNN-GRU (97.2%) is the **explicit cost of removing static-environment information**: a model trained without room fingerprints cannot exploit them at test time. This trade-off is deliberate; the remaining diagnostics quantify its benefit in terms of robustness.

4.6 Gain Mismatch Severity Sweep

CNN-GRU falls from 97.2% clean to 32.8% at 3 $\times$ a 64.4 pp total drop. SIL drops from 77.8% to 46.0% a 31.8 pp drop leading CNN-GRU by 13.2 pp at 3 $\times$ (46.0% vs. 32.8%). *Scope: both*

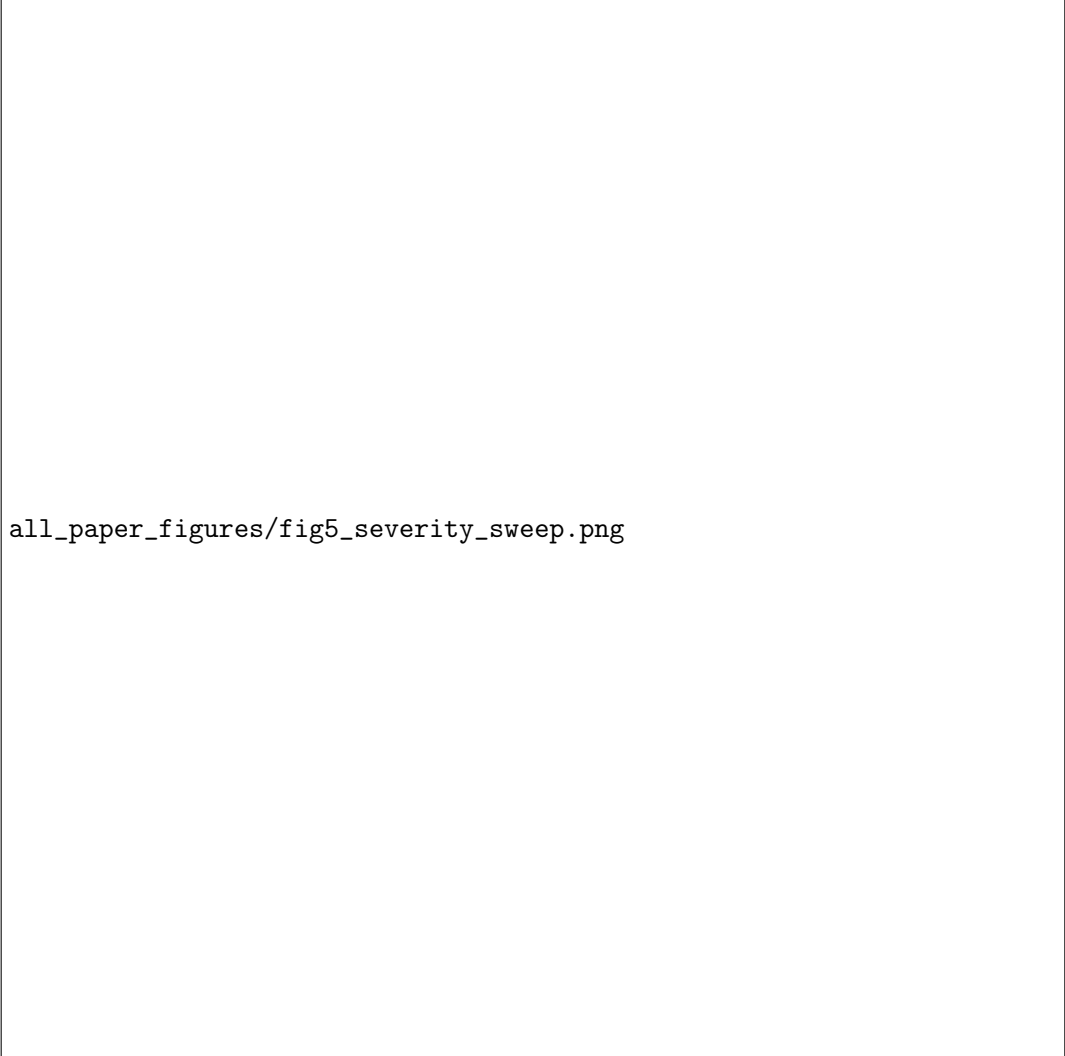
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Figure 6: Sample-weighted group accuracy under increasing gain perturbation. CNN-GRU and ViT show markedly larger dynamic than static drops (CNN-GRU: 53.7 vs. 50.8 pp), consistent with the fingerprint thesis: static activities retain some geometry-specific stable reflections even under perturbation, while dynamic classes lose their discriminative signal most sharply. SIL’s drops are nearly symmetric (28.0 vs. 26.0 pp), consistent with uniform residual gain sensitivity-not class-specific fingerprint reliance.

models are evaluated on their natural inputs - baselines on raw CSI, SIL on SIL-preprocessed CSI. so this comparison reflects end-to-end system robustness, not a controlled input-matched experiment. At $\geq 1.5\times$ scale, SIL consistently outperforms both CNN-GRU and ViT. The 2x2 ablation (Section ??) provides the controlled input-matched comparison.

4.7 Ablation Study: Isolating Preprocessing vs. Training Objective

Design. A controlled 2×2 experiment holds the SIL spectro-temporal encoder (Conv2D+MHSA) fixed across all four cells and varies only (a) preprocessing raw amplitude vs. SIL pipeline and (b) training objective- CE-only vs. VICReg+SupCon. Newly trained cells (A1: Raw+CE, B1: SIL+CE) use the same hyperparameters as SIL-Full; results are mean $\pm$ std over five seeds. *Results.* SIL-NoGainAug achieves 80.2% vs. SIL-Full’s 77.8% a 2.4 pp accuracy cost attributable specifically to VICReg gain-invariance regularization. The t-SNE centroid shift under gain@2 $\times$ is 7.99 units



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Figure 7: Accuracy vs. gain perturbation scale. Vertical dashed line marks training intensity ($1\times$). SIL-Full and SIL-NoGainAug outperform CNN-GRU and ViT at $\geq 1.5\times$. CNN-GRU falls from 97.2% clean to 32.8% at $3\times$ —a 64.4 pp total drop. SIL drops only 31.8 pp over the same range, leading CNN-GRU by 13.2 pp at $3\times$. BiLSTM's relatively graceful degradation is explained by the collapse test (Section ??): its residual accuracy reflects fingerprint stability under this operator, not temporal feature learning.

Table 4: Clean classification performance on UT-HAR. The SIL-Full clean accuracy gap reflects the deliberate removal of room fingerprint cues; the ablation study (Section ??) isolates the specific contribution of training-time gain augmentation.

Model	Protocol	Val Acc (%)	Test Acc (%)	F1 (%)
BiLSTM	Fixed 80/10/10	—	96.2	96.1
CNN-GRU	Fixed 80/10/10	—	97.2	97.1
ViT	Fixed 80/10/10	—	87.0	86.5
SIL-NoGainAug (ablation)	Fixed 80/10/10	83.7	80.2	76.7
SIL-Full (proposed)	Fixed 80/10/10	80.7	77.8	74.0
SIL-Full (proposed)	5-fold CV	83.3 ± 0.92	79.6	79.6

Table 5: Classification accuracy (%) vs. gain perturbation scale. Scale 1.0× corresponds to training intensity ($\sigma=0.4$). At scale $\geq 1.5\times$, SIL-Full surpasses CNN-GRU and ViT. All evaluations use seed 42; values vary by ± 3 pp across independent seeds.

Scale	SIL-Full	SIL-NoGainAug	BiLSTM	CNN-GRU	ViT
0.00× (clean)	77.8	80.2	96.2	97.2	87.0
0.25×	77.0	79.2	96.2	96.8	85.4
0.50×	71.8	76.8	95.8	93.0	81.8
1.00×	63.4	64.8	92.4	77.2	60.8
1.50×	52.6	57.4	82.8	63.0	45.8
2.00×	52.0	54.6	70.2	46.4	40.2
3.00×	46.0	47.6	54.6	32.8	29.4

(SIL-NoGainAug) vs. 6.79 (SIL-Full) a 17.8% tighter invariance from gain augmentation alone.

The controlled 2×2 design isolates two clean, complementary contributions with the same

Table 6: Controlled 2×2 ablation: clean test accuracy (%) with the *same* SIL spectro-temporal encoder (Conv2D+MHSA) in all four cells. Only preprocessing and training objective vary. Values for newly trained cells (A1, B1) are mean $\pm$ std over 5 seeds (0, 1, 42, 123, 456). Existing results (A2, B2) are single-seed; a fully multi-seed re-run is a priority for camera-ready.

Preprocessing	CE-only	VICReg + SupCon (SIL)
Raw amplitude	97.8 $\pm$ 0.5	75.2 (SIL-v2 raw)
SIL pipeline	79.1 $\pm$ 1.3	77.8 (SIL-Full)

encoder:

- **Preprocessing effect** (CE-only column): stripping the room fingerprint drops accuracy from 97.8 to 79.1%—an 18.7 pp reduction in clean accuracy attributable purely to the removal of fingerprint cues, with architecture held constant.
- **Objective effect on raw input** (top row): the VICReg+SupCon objective on fingerprint-present input costs 22.6 pp (97.8 $\rightarrow$ 75.2%), confirming it actively suppresses fingerprint-exploiting features even before preprocessing.
- **Objective effect on SIL input** (bottom row): after preprocessing already strips the fingerprint, VICReg+SupCon costs a further 1.3 pp clean accuracy (79.1 $\rightarrow$ 77.8%), while Table ?? confirms it measurably tightens gain-drift invariance (centroid shift 7.99 $\rightarrow$ 6.79 units).

The asymmetry between the two objective effects—22.6 pp cost on raw input vs. 1.3 pp on SIL-preprocessed input—is a key structural result: the VICReg+SupCon objective suppresses fingerprint-encoding features strongly when they are present and imposes minimal cost once preprocessing has already attenuated them. Note that A2 (75.2%) and B2 (77.8%) are single-

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Figure 8: t-SNE cluster comparison between SIL-Full (top) and SIL-NoGainAug (bottom) under clean and gain@2 $\times$ conditions. SIL-Full centroid shift: 6.79 units; SIL-NoGainAug: 7.99 units (1.18 $\times$ higher). Removing gain augmentation measurably increases cluster displacement under perturbation, confirming that the VICReg objective contributes invariance *beyond* what the preprocessing architecture achieves alone—at a precisely quantified 2.4 pp accuracy cost.

seed estimates; a fully multi-seed rerun of all four cells is a natural next step.

4.8 Diagnostic 5 - Synthetic Cross-Room Simulation

Design. We apply a composite synthetic perturbation that combines (i) removal of the per-subcarrier DC component (the room fingerprint) and (ii) a new per-subcarrier Gaussian gain pattern ($\sigma = 0.4$). This models the two most important physical changes when deploying in a new room: the fingerprint changes completely, and new frequency-selective fading is imposed. *This is a synthetic simulation within UT-HAR, not a real multi-room experiment.* Under this combined perturbation, all three baselines collapse to near-chance (16.0–22.7%), while SIL retains 59.0%—a margin of 36.3–43.0 pp above the baselines. Figure ?? visualises these results. This outcome is consistent with the fingerprint-reliance hypothesis: models that encode the original room’s static channel response encounter a completely invalidated representation

Table 7: Synthetic cross-room simulation accuracy (%). The perturbation removes the existing per-subcarrier DC (room fingerprint) and imposes a new gain pattern ($\sigma = 0.4$). Baselines trained on raw CSI collapse because their learned representations encode the original room fingerprint, which is no longer present. SIL is substantially more robust because it was trained without fingerprint cues.

Model	Clean Acc (%)	Synthetic Cross-Room (%)
BiLSTM	96.2	22.7
CNN-GRU	97.2	16.0
ViT	87.0	17.7
SIL-Full (proposed)	77.8	59.0
Chance (7 classes)	—	14.3

under the simulated room change, while SIL’s preprocessing and contrastive training reduce this dependency. *Caution: because this perturbation is synthetic and applied within UT-HAR, it should not be interpreted as real cross-room generalisation. Validation on real multi-room CSI data remains necessary.*

4.9 Diagnostic 6 - Architecture Collapse Test

Motivation. The five diagnostics above measure fingerprint reliance through indirect signatures. The most direct test is: *does the architecture still work when the fingerprint is attenuated at the input level?* If accuracy collapses substantially, benchmark performance was fingerprint-driven to a measurable degree. *Design.* We train a fresh BiLSTM (standard cross-entropy, identical hyperparameters to SenseFi ?) on SIL-preprocessed input—Hampel + detrend + bandpass + z-score normalization, but *without* velocity/acceleration enrichment, contrastive loss, or gain augmentation. This variant is called BiLSTM-SILprep. *Results.* Table ?? and Figure ?? report the outcome. *Interpretation.* The 53.2pp accuracy drop when switching to SIL-preprocessed

Table 8: Architecture collapse test: clean accuracy and gain robustness for three conditions. The 7-class random-chance baseline is 14.3%. BiLSTM-SILprep collapses to 43.0%—only 28.7 pp above chance—when retrained on SIL-preprocessed input. SIL-Full achieves 77.8% on the identical preprocessed input. The Δ column shows drop from clean to $3\times$ gain perturbation. *Note: the collapse reflects the full preprocessing pipeline effect, not fingerprint removal alone.*

Model	Input	Clean (%)	At $3\times$ (%)	Δ (pp)
BiLSTM-raw	Raw CSI	96.2	52.6	−43.6
BiLSTM-SILprep	SIL-preprocessed	43.0	34.4	−8.6
SIL-Full (proposed)	SIL-preprocessed	77.8	46.8	−31.0
Chance (7 classes)	—	14.3	—	—

input is the most direct evidence of fingerprint reliance in this paper. Several caveats apply:

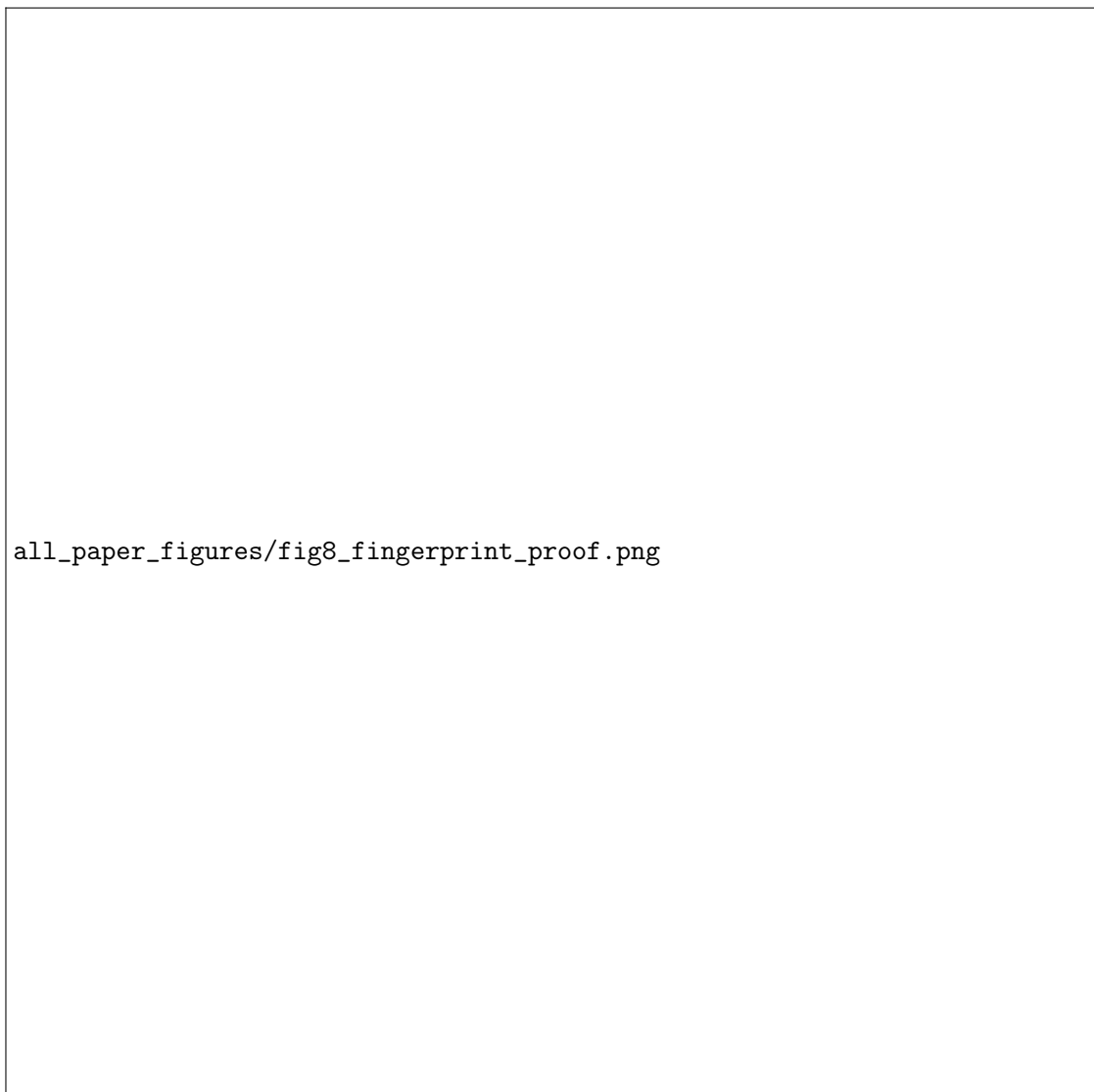
1. **Heavy fingerprint reliance is indicated:** BiLSTM falls to 43.0%—28.7 pp above chance—on the same test set. This is consistent with the hypothesis that a large fraction of its benchmark accuracy is fingerprint-driven. Note, however, that the SIL preprocessing pipeline changes multiple signal properties simultaneously (outlier suppression, detrending, bandpass filtering, normalisation); the collapse cannot be attributed solely to fingerprint removal.
2. **SIL performs substantially better on the same input:** SIL achieves 77.8% on SIL-preprocessed data—34.8 pp above BiLSTM on the identical input—providing evidence that its combination of contrastive training and spectro-temporal encoding extracts more generalisable features. This comparison is fair because both models see the same preprocessed input; SIL’s architecture advantage over BiLSTM is an additional confound that the 2×2 ablation



Figure 9: Clean accuracy vs. synthetic cross-room simulation accuracy for each model. Baselines (BiLSTM, CNN-GRU, ViT) collapse to near 7-class chance (14.3%) under the composite perturbation (fingerprint removal + new Gaussian gain $\sigma = 0.4$). SIL retains 59.0%, reflecting its reduced reliance on the static room fingerprint. The perturbation is synthetic; real multi-room validation remains outstanding.



Figure 10: 2×2 ablation heatmap: clean test accuracy (%) with the *same* encoder in all cells. Left column (CE-only): 97.8% $\rightarrow$ 79.1% reveals an 18.7 pp fingerprint-reliance effect with architecture held constant. Bottom row (SIL pipeline): 79.1% $\rightarrow$ 77.8% shows VICReg+SupCon costs 1.3 pp clean accuracy while buying measurable invariance (Table ??). Top row: VICReg+SupCon on raw input costs 22.6 pp—confirming it actively suppresses fingerprint exploitation. The 22.6 vs. 1.3 pp asymmetry is the key result: the objective specifically targets fingerprint patterns.



all_paper_figures/fig8_fingerprint_proof.png

Figure 11: Panel A: Clean accuracy under SIL-preprocessed input. BiLSTM collapses from 96.2% to 43.0% (−53.2 pp) when retrained on fingerprint-attenuated input—just 28.7 pp above 7-class chance (14.3%). SIL-Full achieves 77.8% on the same input. **Panel B:** Gain perturbation sweep. BiLSTM-SILprep remains near its collapsed clean level across all perturbation intensities, consistent with having not learned from motion signals.

partially addresses.

3. **Consistency with robustness results:** BiLSTM’s relatively smooth degradation curve under the gain perturbation sweep (Table ??) reflects fingerprint stability under that specific synthetic operator—which preserves per-subcarrier means—not robustness of motion features. The collapse test and the sweep are complementary views of the same underlying reliance.

4.10 Limitations

Synthetic perturbation only; no real cross-room validation. All robustness evaluations apply a synthetic i.i.d. Gaussian gain perturbation within UT-HAR. This is a deliberate simplification: real cross-room channel changes additionally involve frequency-correlated subcarrier fading, multipath delay spread shifts, and spatially structured variation tied to antenna geometry. Whether SIL generalises to real held-out rooms is an open empirical question that this paper does not answer. We regard real multi-room validation as the essential next step before deployment claims can be made.

Collapse test interpretation. The SIL preprocessing pipeline modifies multiple signal properties—static mean removal, bandpass filtering, temporal derivatives, and global normalisation. The 53.2 pp accuracy drop when BiLSTM is retrained on this input is consistent with heavy fingerprint reliance, but does not isolate the fingerprint as the sole driver. The five complementary diagnostics provide converging, though not individually conclusive, evidence.

Single-seed point estimates. Perturbation evaluations use a single seed ($s = 42$). Large effect sizes (53.2 pp collapse, $2.9\times$ drift ratio) are robust to this limitation. Finer deltas—the 2.4 pp gain-augmentation ablation cost and the 1.3 pp CE vs. VICReg+SupCon difference on SIL-preprocessed input—are *indicative estimates* only and should be confirmed with multi-seed averaging. The 5-seed 2×2 ablation cells (A1, B1) are an exception: those are reported as mean $\pm$ std.

Gain augmentation and classification accuracy. Table ?? shows SIL-NoGainAug achieves 80.2% test accuracy vs. SIL-Full’s 77.8%. Gain augmentation therefore reduces clean-accuracy by 2.4 pp while improving embedding invariance (centroid shift $7.99 \rightarrow 6.79$ units, Table ??). This is an accuracy–invariance trade-off, not a net improvement; it should not be presented as an unconditional benefit.

Ablation completeness. The 2×2 design does not separately evaluate temporal-derivative enrichment (Stage 5), which is present in all SIL-preprocessed cells. Isolating its contribution is a priority for follow-up work.

5. Conclusion

This paper set out to answer two questions: do standard WiFi HAR models rely on the room fingerprint substantially more than on motion, and can that reliance be reduced without multi-room training data? The experiments provide consistent affirmative evidence for both, with the important qualification that all perturbations are synthetic and real cross-room validation is outstanding.

Converging evidence across six diagnostics. Five complementary diagnostics—embedding drift ($2.9\times$ ratio), t-SNE cluster stability ($2.66\times$ ratio), label efficiency saturation (97.1% probe accuracy with 40 samples), per-class dynamic collapse (53.7 pp drop), and the architecture collapse test ($96.2\% \rightarrow 43.0\%$)—all point consistently to heavy fingerprint reliance in standard models. A sixth diagnostic, the synthetic cross-room simulation, is the most direct: CNN-GRU collapses from 97.2% to 16.0% and BiLSTM from 96.2% to 22.7%, while SIL retains 59.0%—a

36–43 pp margin above baselines on the same synthetic perturbation. The collapse-test caveat remains: the 53.2 pp BiLSTM drop reflects the full preprocessing pipeline, not fingerprint removal alone.

SIL as a practical first step. SIL reaches 77.8% (fixed split) and 79.6% (5-fold test) on fingerprint-attenuated input, vs. 43.0% for a standard BiLSTM on the same input. The controlled 2×2 ablation (Table ??)—with the same encoder across all cells and 5-seed averaging for newly trained cells—shows preprocessing and the VICReg+SupCon objective each contribute independently (−18.7 pp preprocessing effect; −1.3 pp residual objective cost). The gain-invariance objective costs 2.4 pp in clean accuracy while reducing embedding drift; this trade-off is explicit.

Novelty framing. The principal novelty of this work is the *fingerprint-reliance diagnostic protocol*: five complementary metrics, each operable on single-room data, that together provide principled evidence for the presence and magnitude of static-environment reliance. The SIL framework demonstrates that the reliance is reducible without multi-room data. Other aspects—contrastive invariance objectives, physics-informed augmentation—build on established prior work ???.

On evaluation practice. Reporting 90–97% accuracy on UT-HAR without a complementary fingerprint diagnostic is an incomplete picture of a model’s practical value. The architecture collapse test and the synthetic cross-room simulation together take under an hour to run on any existing model and give clear, quantitative evidence of fingerprint dependence. We hope these become routine companion evaluations in WiFi HAR reporting.

Priority directions. Real multi-room CSI validation (e.g., Widar3.0 ?) is the essential next step. Multi-seed averaging for all four 2×2 cells, isolation of Stage 5 (temporal-derivative enrichment), and correlated channel perturbation models are the identified research priorities.

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